

Short Questions

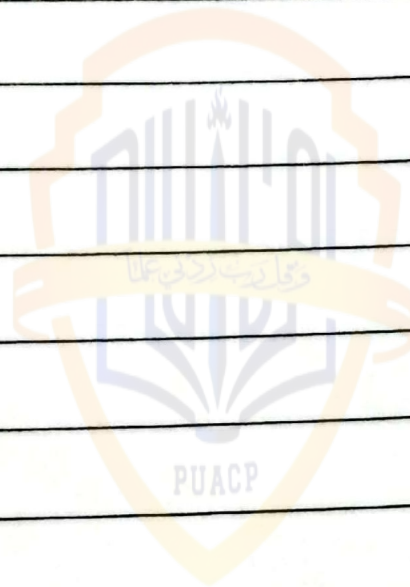
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Preference Choice Under Uncertainty

Good Luck



Q#1

Pareto Improvement

Pareto Improvement named after the Italian economist Vilfredo Pareto, is a concept in econo. that describes a situation where one or more individuals can be made better off without making anyone else worse off. It's a way to evaluate changes in allocation of resources to efficiency and fairness.

Definition:

A Pareto improvement occurs when a change in allocation benefits at least one person without causing any harm or disadvantage to others.

Factors / Key Aspects:

- Resource Allocation
- Trade
- Pollution Reduction

- (1) **Benefit without harm:** At least one party benefits, and no party is worse off.
- (2) **Efficiency:** Indicates a more efficient effort of resources.

Examples:

Scenario: Consider two roommates, Sam and Alex, who share an apartment. Initially, Sam has a larger bedroom, and Alex has a smaller one. Both pay equal rent.

Pareto Improvement: Alex proposes that Sam keeps the larger room but pays slightly more rent, while Alex pays less rent. Sam agrees because the room is worth the extra cost to him, and Alex is happy to pay less rent for the smaller room.

Outcomes:

- Sam better off (larger room, but willing to pay extra)
- Alex Better off (smaller room, but lower rent)
- No One is worse off Both agree to the new terms, making this a Pareto improvement.

Importance:

Pareto improvements are valuable in economic theory because they help to identify changes that led to more efficient and equitable outcomes.

Q#02

Nash Equilibrium

Definition:

Nash equilibrium is a key concept in game theory, named after the mathematician John Nash. It describes a situation

in a strategic game where no player can benefit by unilaterally changing their strategy, given that the other players' strategies remain unchanged. In other words, each player's strategy is optimal when considering the decisions of other players.

At Nash equilibrium, every player is doing the best they can given the strategies of others.

Example:

Prisoner's Dilemma: Two criminals are arrested and interrogated separately. They have the following choices:

- Cooperate (Stay Silent)
- Defect (betray the other)

The possible outcomes are:

- If both cooperate, they each

- get a light sentence (1 year).
- If one defect and the other cooperates, the defector goes free and the cooperator gets a heavy sentence (10 years).
 - If both defect, they both get a moderate sentence (5 years).

Payoff Matrix

	Cooperate (B)	Defect (B)
Cooperate (A)	$(-1, -1)$	$(-10, 0)$
Defect (A)	$(0, -10)$	$(-5, -5)$

In this game, the Nash equilibrium occurs when, both the both players choose to defect. Despite the fact that mutual cooperation would lead to a better collective outcome, defecting is the best response for both players when considering the strategy of the other.

Q#03 Payoff Matrix

Definition:

A payoff matrix is a table that represents the payoffs or outcomes of different strategies chosen by players in a strategic game. It shows the rewards or penalties that each player receives depending on their own and the other player's action.

Payoff matrix are commonly used in game theory to analyze competitive situations where the outcomes for each participant depends on the choices made by all participants.

Example:

Same exple discussed in Q#02.

Q #04

Pareto Efficiency

Definition:

Pareto efficiency also known as Pareto optimality, is a concept in econo. that describe a state of resource allocation where it is impossible to make any individual better off without making any one individual worse off. It represents an optimal distribution of resources from which no further mutual gains can be made.

Key Characteristics

- **Efficiency:** All resources are allocated in a more efficient manner.
- **No Wasted Resources:** No resources can be reallocated to improve someone's situation without hurting someone else.

Equilibrium: The system is in equilibrium

where no further Pareto improvements (beneficial reallocation) can be made.

Example:

Scenario: Imagine an economy with two goods A & B, and two consumers, X & Y.

• Initial Allocation

- X has 3 units of A & 2 units of B.
- Y has 2 units of A & 3 units of B.

• Redistribution Possibility

- If we attempt to reallocate resources, giving X more units of A, Y would lose some units of A and vice versa.

Overview:

A Pareto Efficiency, we cannot reallocate that good to make one consumer better off without making the other worse off. In this case, we might find that the given allocation (e.g. X, with 3A & 2B, Y with 2A & 3B) cannot be altered

to make one better off without disadvantaging the others thus achieving Pareto efficiency.

Q#05

State Moral Hazard

Definition:

Moral Hazard is a technique in which one party engages in riskier behaviour or fails to act in good faith because they do not bear the full consequences of their actions.

This often occurs when one party is protected from risk, typically through insurance or other guarantees, leading to an incentive to act in ways that are not in the best interest of the other party involved.

Key Characteristics:

- (1) **Asymmetric Information:** The party taking the risk has more info about their behavior and the risk involved than the party bearing the consequences.
- (2) **Incentive Problem:** The party shielded from risk might take actions that increase the likelihood or severity of negative outcomes because they do not fully face the repercussions.

Example:

Insurance

- **Scenario:** An individual has a comprehensive health insurance policy.
- **Behavior:** Knowing that their medical expenses will be covered, they might be less concerned about engaging in healthy behavior or might overuse medical services.

- **Consequences:** The insurance company faces higher costs due to the increased risk and overuse of medical services.

Overview:

Moral Hazard emphasize the importance of aligning incentives so that individuals and organizations face the true costs of their actions.

This alignment helps promote responsible behavior and reduces the likelihood of excessive risk taking.

Q#6

Free Rider Problems

Definition:

The free rider problems occurs when individuals or entities benefit from resources, goods or services

without paying for them, relying instead on others to bear the cost. This leads to under-provision of those goods or services because people have an incentive to avoid paying for what they can enjoy for free. The issue is commonly associated with public goods, which are non-excludable (can't prevent anyone from using them) and non-rivalrous (one person's use doesn't reduce availability) to others).

Key Characteristics

- 1- **Non-Excludable:** People cannot be prevented from using the goods or services.
- 2- **Non-Rivalrous:** One person's use doesn't diminishes the availability to others.
- 3- **Incentive To Avoid Paying:** Individuals

Can benefit without contributing to the cost.

Example:

Public Goods

- **Scenario:** Consider a city park that is open to everyone.
- **Behavior:** Residents use the park for leisure, exercise and socializing, but not everyone contributes to its maintenance through taxes or donations.
- **Consequence:** If too many people act as free riders, the park may become underfunded and deteriorate, reducing its quality and benefits for everyone.

Q#07

State Risk Diversification.

(Ans)

Risk Diversification

Def:

It is a strategy to reduce the overall risk of investments by spreading them across various assets. Instead of putting all your eggs in one basket, you place them in multiple baskets. This way, the poor performance of one investment won't heavily impact your overall portfolio.

Example:

Imagine you invest all your money in stocks of one company. If that company performs poorly, you could lose a significant portion of your investment.

However, if you diversify your investments by putting some money in different stocks, bonds and real estates, the risk is spread out. Even if one investment performs poorly, others might perform well, balancing out the overall risk.

Q#08

First fundamental Theorem of welfare Economics.

(Ans)

Fundamental Theorem:

Every competitive equilibrium leads to a Pareto Efficient allocation of resources.

Explanation

In a perfectly competitive

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market, where no single buyer or seller can influence prices, and all parties have complete information, the market forces naturally guide the allocation of resources in such a way that it's impossible to make any one person better off without making someone else worse off.

Example

Consider a farmers' market. If everyone buys and sells freely, the prices of goods adjust to so that everything produced is sold, and everyone's needs and wants are satisfied. If the price of apples is too high, fewer people buy them, leaving excess apples. If the price is too low, apples sell out quickly. At the right price, the supply

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and demand balance, leading to no leftover apples and no unmet demand - a state of Pareto efficiency.

Q#09

Walras' Law

(Ans)

WALRAS' LAW:

In an economy, with multiple markets, the sum of excess demands must equal the sum of the excess supplies. In other words, the general equilibrium, if all but one of the markets are in balance (where supply equals demand), the last market must also clear. This ensures that all markets in the economy are

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In equilibrium.

The law implies that it's enough to analyze $n-1$ markets in an n -market system to determine overall equilibrium.

Example

⇒ Scenario:

- Market 1: Apples
- Market 2: Oranges
- Market 3: Bananas

⇒ Given

Suppose we know the following:

In the apple market, the quantity supplied perfectly matches the quantity demanded, so there's no excess supply or demand.

In the orange market, the quantity supplied also equals the quantity demanded, so no excess supply or demand here either.

Walras' Law Applied:

According to Walras' Law, if the sum of excess demands across all the markets must be zero, then:

If $E_1 = 0$ (excess demand in apple), and $E_2 = 0$ (excess demand in orange), then the excess demand in banana market (E_3) must also be zero.

Conclusion:

So, if the markets for apples and oranges are both in equilibrium the banana market must also be in equilibrium.

This exemplifies how Walras' Law ensures that total excess demand is zero when all but one of the market are in equilibrium.

O#10

Non-Cooperative & Cooperative Games

(Ans)

Non Cooperative Games:

In non-cooperative games, each player acts independently, often in their own self-interest, and without collaboration or communication with other players. The focus is on individuals' strategies and the resulting payoffs.

Characteristics:

Here are the following characteristics of Non-cooperative games:

- Individual decision Making
- Strategic Interaction
- Nash Equilibrium

Examples:

(1) Prisoner's Dilemma: Two prisoners

must decide whether to confess
or remain silent without knowing
the choice of the other.

(2) **Rock-Paper-Scissors:** Each player
independently chooses one of the
three options to outperform the
other.

Cooperative Games:

In cooperative games, players
can perform coalition, collaboration,
negotiate to achieve better outcomes.
In fact, it is on joint strategies,
and how the total payoff can
be distributed among the players.

Characteristics:

- Coalitions & Collaboration
- Joint Strategies
- Fair Supply Value

Examples:

(1) **Prisoner's Dilemma:** Two prisoners

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negotiate trade agreements to benefit both parties.

- (2) **Joint Ventures:** Companies collaborate to develop the new product, agreeing on how to share profits.

Summary

Aspects	Non-Cooperative Games	Cooperative games
Player Interaction	Independent	Collaborative
Strategic Info.	Individual	Joint
Communication	None / Limited	Allowed / Integral
Outcome focus	Indi. Payoff maxim.	Joint Payoffs & Fair Dist.
Examples	Prisoner's Dilemma, RPS	Bargaining, Joint Ventures.

Q#11

Edgeworth Box.

(Ans)

The edgeworth box is a fundamental

concept in microeconomics, used to represent various distribution of resources and the efficiency of allocations between two parties.

Edgeworth Box

(1) Purpose:

Illustrate the benefits of trade, showing how two parties can achieve mutual benefit.

(2) Structure:

A rectangular diagram where each axis represents the total amount of a good variable in an economy.

(3) Points:

Each point within the box represents the different allocation of resources between two individuals.

(4) Utility:

Helps in understanding concepts

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like Pareto efficiency, where no one can be better off without making someone else worse off.

Example

Imagine you and a friend have some amount of two different candies. The box helps visualize all the ways you could share those candies, showing when neither of you can be better off without the other being worse off.

Q #12

Measures for free Rider Problem

(Ans)

The free rider problem occurs when individuals or entities benefit from resources, goods

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or services without paying for them, leading to under-provision of those goods or services.

Here are some measures to mitigate the free riders problem.

Measures

Govt. Intervention: Implement taxes or regulations to fund public goods.

Private Provision: Encourage voluntary contributions and create excludable goods.

Clubs & Memberships: Use membership fees to provide exclusive benefits.

Public-Private Partnership: Form joint ventures between govt. and private firms.

Social Norms & Peer Pressure: Promote community enforcement and peer monitoring.

Q#13

State Risk Spreading.

(Ans)

Risk Spreading

Definition:

Risk Spreading refers to the practice of distributing risk across multiple parties or areas to reduce the exposure to any single source of potential loss. This concept is widely used in financial investments, insurance, and business strategies.

Here are a few ways it's implemented:

Diversification

Investing a variety of assets to reduce the impact of any one asset's poor performance.

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Insurance

Transferring risk, foreign insurance by paying premiums.

Partnership & Joint Ventures

Collaborating with other businesses to share risks associated with large projects.

Supply Chain Management

Using multiple suppliers to avoid dependency on a single source.

Q#14

State Asymmetric Information.

(Ans)

Asymmetric Information

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Definition:

It refers to a situation where one party in a transaction has more or better information compared to the other. This imbalance can lead to various economic problems and inefficiencies.

Key Aspects

Adverse Selection: Occurs when buyers and sellers have access to different information, leading high-quality products or reliable individuals to be undervalued or driven out of the market.

Moral Hazard: Arises when one party takes on more risk because they do not bear the full consequences of that risk, knowing that another party will be incurring the cost.

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Examples:

Insurances and Credit market helps making the suitable suggestions for Asymmetric Information.

Solutions

- **Screening & Testing:** Methods like background checks, health exams and credit scores to reduce information Asymmetric.
 - **Incentive Schemes:** Aligning the interests of both parties to promote more optimal behavior.
 - **Regulation & Disclosure Requirements:** Enforcing rules that require the sharing of relevant information.
-

Q#15

Symmetric Information.

(Ans)

Symmetric Info-

Definition:

It refers to a situation where all parties involved in a transaction have access to the same info.

This balance of knowledge helps ensure fair and efficient markets.

Key Aspects

- **Transparency:** Both buyers and sellers have full knowledge of the product or service being exchanged.
- **Fair Transaction:** With equal info, the likelihood of unfair practices or pricing is minimized.
- **Market Efficiency:** Symmetric information

can lead to more efficient markets as all participants make informed decisions.

Examples

- (1) **Stock Markets: Ideal Scenario.** where all investors have access to the same financial statements and market data.
- (2) **Online Marketplace:** Platforms where sellers provide detailed product descriptions and buyers can review and compare options before making a purchase.

Ensuring Symmetric Info.

- **Disclosure Requirements:** Regulations that mandate the sharing of key information with all stakeholders.
- **Independent Audits:** Third party audits to ensure info accuracy and transparency.
- **Technology:** Using digital platforms.

Q#16

State the Risk Premium.

(Ans)

Risk PremiumDefinition:

The risk premium is the additional return that an investor requires to compensate for the higher risk of a particular investment compared to high risk-free asset. It reflects the difference between the expected return on the risky asset and the risk-free rate.

Key Points

- (1) **Risk Compensation:** The risk premium compensates investors for the possibility that the actual return may be lower than the expected return.

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(2) **Market Risk:** Higher risk investments typically offers higher potential returns to attract investors unwilling to take on extra risk without extra reward.

(3) **Types Of Risk:**

These can include:

- Market Risk
- Credit Risk
- & Other Uncertainties.

Calculation

$$\text{Risk Premium} = \text{Expected Return} + \text{Risk-Free Rate}$$

Example

If a stock has an expected return of 8% and the risk-free rate (like: govt. bond rate) is 2%, the risk premium would be 6%.

Q#17

Define Coase Theorem.

(Ans)

Coase Theorem

Definition:

The Coase theorem, formulated by economist **Ronald Coase**, is a fundamental concept in law and economics. It states that if property rights are well-defined and transaction costs are negligible, parties will negotiate to correct externalities and allocate resources efficiently, regardless of the initial distribution of property rights.

Key Element

- (1) **Property Rights:** Clear and enforceable property rights are essential for the theorem to hold.

(2) **Transaction Costs:** The theorem assumes that parties can bargain without incurring significant costs.

(3) **Efficient Allocation:** Parties will ^{and/or} negotiate to reach an outcome that maximizes total welfare, correcting any market failures caused by externalities.

Example.

Imagine a farmer and a nearby factory. The factory's emission harms the farmer's crops. According to the Coase Theorem, if they can negotiate without high transaction costs, they can agree on a solution, such as the factory paying the farmer or investing in pollution control, to minimize damage and improve overall efficiency.

Q#18

Define Subgame Perfect Equilibrium.

(Ans)

Sub Game Perfect Equil.

Definition:

Subgame Perfect Equilibrium (SPE) is a defined concept in game theory. It represents a strategy profile in extensive form games where players' strategies provide optimal outcomes for every possible subgame of the original game.

In simpler terms, it ensures players make the best decisions at every stage of the game, considering all possible future actions.

Key Points

- (1) **Extensive Form - Game:** Typically applied to games represented in a tree structure, where sequences of decisions and strategies are depicted.
- (2) **Optimal Strategies:** SPE strategies result in the best possible outcomes at each decision point within the game.
- (3) **Backward Induction:** A common method used to find SPE by starting from the end of the game and reasoning backward to determine optimal strategies.

Example

Imagine a chess game where each player aims to make the best possible move not only in response to the current position but also considering all future potential.

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moves by both players.

Using SPE, players would plan their strategies to ensure the best outcome from every point in the game, ensuring no mistakes at any stage.

Q#19

Define Adverse Selection.

(Ans)

Adverse Selection

Definition:

Adverse Selection is a concept in economics and insurance that occurs when there is an imbalance of information between buyers and sellers. It happens when one party, typically

the buyer, has more information about the product or service than the party, leading to suboptimal market outcomes.

Key Points

- (1) **Imbalance of Information:** One side has more or better information, leading to mismatched expectations.
- (2) **High-Risk Individual:** Those who are more likely to use the service or product tend to participate more, increasing the risk for providers.

Example

Insurance Market:

High risk - individuals (like those with health issues) are more likely to buy health services/insurance, while low risk individuals might avoid it. This leads insurers to raise

premium to cover potential losses, making insurance less affordable and accessible for everyone.

Solutions

- **Screening:** Techniques like medical exams for health insurance or credit checks for loans to evaluate and manage risk.
- **Incentive Alignment:** Designing policies or contracts that encourage the honest disclosure of information.

Q#20

Define Dominant Strategies.

(Ans)

Dominant Strategies

Definition:

It is a concept in game

theory where a player's best strategy is independent of what their opponent does. This means that no matter what the other players choose, the dominant strategy always provides the highest payoff for the player who adopts it.

Key Points

- (1) **Independence:** The choice of a dominant strategy does not rely on the action of others.
- (2) **Optimal Outcomes:** It's the best possible choice, yielding the maximum benefit for the players.
- (3) **Game Theory:** Often used to analyze the situations in competitive environments like markets, politics and more.

Example

Imagine a classical game of Rock, paper, Scissors. If "Rock" somehow always guaranteed a win regardless of the opponent's choice, choosing "Rock" would be the dominant strategy.

But in real world, dominant strategies are rare. Often players must consider the potential of their opponents, leading to equilibria like the Nash Equilibrium instead.

Q # 21

State the preference Choice under Uncertainty.

(Ans)

Preference Choice (Under Uncertainty)

Definition:

It refers to how individuals make decisions when the outcome are not certain. This field is particularly important in econo, finance and decision theory.

Key Aspects

- (1) **Utility Theory:** Individuals assign a value of satisfaction (utility) to different outcomes and choose the option that maximize their expected utility.
- (2) **Risk Aversion:** The tendency to prefer options with more certain outcomes over those with higher but uncertain returns.
- (3) **Prospect Theory:** Describes how people make choices based on the potential value of losses and

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gains, often weighing losses more heavily than equivalent gains.

Example

Investment Choices:

When choosing between a guaranteed return (like a fixed deposit) and a potential high return with high risk (like stocks) and investors evaluate the utility of possible outcomes, factoring in their risk tolerance.